My Data Literacy Project
(Replace this with your Project Title)

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Abstract

Put your abstract here. Abstracts typically start with a sentence motivating why the subject is interesting. Then mention the data, methodology or methods you are working with, and describe results.

1. Introduction

As the world's largest athletics association, the Deutscher Leichtathletik-Verband (DLV) is tasked with facilitating peak individual performance and maintaining an internationally competitive national team (Deutscher Leichtathletik-Verband, n.d.). However, zero medals at the 2023 World Athletics Championships in Budapest and a historic low medal count at the 2024 Paris Olympics have left the organization in a sobering position. Although individual athletes still occasionally reach the podium, the declining return on investment in terms of Olympic success (**?**) raises questions about the underlying health of the national talent pool.

While medals are the most visible metric of success, they reflect only the peak of elite performance and offer little insight into how performance is distributed within the elite level. To examine potential structural weaknesses, we analyze the top 35 rankings from 2001 to 2025, guided by two research questions. First, we assess the impact of the COVID-19 pandemic as a short-term disruption to the development of elite performances. Secondly, we are investigating whether the performance gap between the national elite and the broader field of competitors has widened and how performance has generally developed in comparison. This investigation is prompted by observations at national championships where a growing decline in performance was noted beyond the top positions. We aim to quantify this reduction in competitive intensity, as a lack of internal pressure, in line with the principle that "competition fosters performance," can lead to top athletes falling behind internationally.

Our analytical approach is detailed in Section 2, where we describe the data processing and justify the analytical framework. Building on this, we will analyze performance metrics to identify structural trends and evaluate our research questions (Section 3). Finally, Section 4 discusses the implications of these findings.

2. Data and Methods

2.1. Data Acquisition and Standardization

The data were retrieved from the official archive by the Deutscher Leichtathletik-Verband (DLV), which documents the top performances across the major athletic disciplines (including running events, hurdles and steeplechase, jumping events, throwing events, and combined events) from 2001 to 2025. The archive covers performances across gender (male, female) and age groups (age 14 up to adults).

As the data for each year, age group, and gender are stored in separate PDF files, we used *pdfplumber* in combination with specific regular expressions to handle formatting differences over years. Outliers due to entry inconsistencies in the original files (e.g., irregular time formats) were corrected when the intended values could be inferred from external sources. Minor data gaps resulting from parsing errors were manually completed by referencing the original PDF files. Non-recurrent distances (e.g., [3000 m steeplechase and] 7.5 km events in selected youth categories) were excluded, yielding a final dataset of 226,683 entries.

We standardized discipline names and birth years to a four-digit format to ensure consistency across the dataset. Performance marks were converted into numeric values (seconds for track events and meters for field events) to allow statistical analyses. To facilitate comparisons across events (e.g., 100m sprint vs. javelin throw), performances were transformed using World Athletics scoring tables (IAAF scores) (Wor). The IAAF scores are a standardization metric maps performance marks onto a common point scale. They are

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released annually by World Athletics as tables in PDF files, but unfortunately there is no public formula to calculate the scores directly from performance marks. However, some people have tried to reverse engineer the formulas for the scoring tables. The one that we use, originates from a blog post by Jeff Chen (Chen, 2019). He parsed the PDF files of the scoring tables and fitted a quadratic function to the data points to get an approximate formula with coefficients for each discipline. We implemented his formulas in our data processing pipeline to convert all performance marks into World Athletics points for comparability across disciplines. In the following, all our calculations and analyses are based on these IAAF scores.

To ensure a robust and representative sample for all statistical evaluations, the dataset included the top 35 national performances in each group and year. This threshold ensures that there is enough data for all groups. A ”Group” is defined as the combination of Gender x Age group x Discipline, maintaining consistency with the original leaderboards. The study focuses on the U18, U20, and Adult age groups, as these represent the most relevant categories, since these age groups compete at international world level championships. To allow for cross-disciplinary comparison, four discipline groups were defined based on current team selection criteria (DLV): sprints, runs, throws, and jumps.

2.2. Corona Heatmap

To investigate the impact of the pandemic on professional athletics, an exploratory data analysis was performed to identify structural and qualitative shifts in performance. To gain an initial understanding of the dataset, a heatmap-based plot was employed. This visualization maps the aggregate frequency of competitive entries across a grid of months and years, offering a clear view of seasonal activity levels and data density.

2.3. Median Analysis

To isolate the specific impact of the pandemic year, the dataset was partitioned into two distinct cohorts. The baseline group, denoted as S_{base}, consists of all pooled IAAF scores recorded between 2015 and 2019. This aggregation serves as a stable reference for ”normal” pre-pandemic performance levels. This analysis was performed to investigate a potential general trend across the investigated disciplines. To evaluate the directionality and magnitude of the performance shift, the relative difference between the group medians ($\tilde{x}$) was calculated. The relative percentage change is defined by the following formula:

$$\Delta_{rel} = \left(\frac{\tilde{x}_{2020}}{\tilde{x}_{base}} - 1 \right) \times 100$$

Through this approach, the exact degree of performance deviation relative to the pre-pandemic baseline was precisely quantified, determining the specific percentage by which median performance decreased or increased. To statistically validate the observed shifts in performance, a Mann-Whitney U test was performed. This non-parametric test was selected because it does not require the assumption of a normal distribution, which is particularly relevant for elite athletic data that often exhibits skewness.

2.4. Density-Analysis

To evaluate the impact of the pandemic on general performance density, we analyzed the dispersion of IAAF scores using a non-parametric bootstrapping approach. This method allows us to determine whether the competitive field became more spread out or remained stable during the 2020 season. The interquartile range (IQR) was chosen as the main measure of dispersion instead of the standard deviation (STD) because it is more robust when analyzing elite athletic data. High-performance scores often include ”outliers,” extraordinary performances that can greatly distort the standard deviation. The primary metric for measuring these changes is the relative IQR difference (θ), which quantifies the percentage change in performance spread in 2020 compared to the pooled baseline from 2015 to 2019. This statistic is defined by the formula:

$$\theta = \left(\frac{IQR_{2020}}{IQR_{base}} - 1 \right) \times 100$$

A positive value indicates an increase in dispersion, suggesting the performance gap between athletes widened, whereas a negative value indicates a shift toward a more homogenous field.

2.5. Performance evolution analysis

To analyze long-term structural changes, we compare the five-year periods 2001–2005 and 2021–2025. This timeframe smooths out annual fluctuations and, through a broader data base, ensures the statistical validity of the results. The analysis is conducted on two levels: 1. Recording performance trends and density for all groups to identify group differences. 2. Examining the development to identify the structure in which performance changes occur. This allows us to identify general trends and determine the structure of the changes.

To assess general changes in performance, the percentage difference between the medians is used, as the median represents the typical performance level and is more robust against outliers compared to the mean. Medians are determined for each group from the data pools of the respective five-year periods. The development is then calculated as a percentage difference, normalized to the baseline period.

A positive value indicates an improvement in performance, while a negative value represents a decline. Statistical significance is tested using the Mann-Whitney U test, which allows for a comparison of the periods without assuming a normal distribution.

To measure the performance density, we define the performance gap (G_p) between the three best and the three worst (e.g., 33rd–35th place) in our data set for the two time periods. The averaging of the top 3 reflects the international standard of success (podium), while the bottom 3 provide a stable and comparable baseline. We used the performance gap to measure the breadth of performance at the top level and utilized all available information (e.g., rankings) from our dataset to measure polarization without assuming a normal distribution. Mathematically, the gap (G_p) is defined as follows:

$$G_p = \left(\frac{1}{3|P|} \sum_{y \in P} \sum_{r=1}^{3} x_{y,r} \right) - \left(\frac{1}{3|P|} \sum_{y \in P} \sum_{r=33}^{35} x_{y,r} \right) \tag{1}$$

where $x_{y,r}$ represents the individual performance of rank r in year y, and $|P|$, in our case $|P| = 5$, represents the number of years per period. The development of the performance gap was calculated as a percentage change between the start and end periods, normalized to the start period. A positive value indicates a decrease and a negative value indicates an increase in the performance gap.

Statistical significance was determined using bootstrap resampling with B = 10,000 iterations. This is a non-parametric method and is well-suited for the small sample size (N = 12) in our case. By drawing with replacement from the performance pools, we generated a distribution of the differences in performance gaps to calculate the 95% confidence interval (CI). If the CI contains zero, the null hypothesis that the performance density does not change significantly cannot be rejected.

To analyze the development of the different rankings, we calculate the arithmetic mean for each rank across all groups for the start and end periods. For each rank, even when differentiating by age group and gender, the sample size is at least n = 80. According to the central limit theorem, we can assume a normal distribution and thus calculate the arithmetic mean. The development of each rank was determined by calculating the percentage change between the start and end periods, normalized to the start period. A positive value indicates an improvement in the average ranking, a negative value a decline.

There were changes in weights and heights in the period of our data. In the U20 category some throwing weights and the height of 110 m hurdles changed (see (Pro)). We observed changes in 2012 in the weights of U18 categories. The results of the median and rank analyses were annulled, as these results are solely attributable to the change. The results of the performance gap analysis were retained, as these are independent of absolute performance levels.

3. Results

3.1. Corona Heatmap

Analysis of the distribution of seasonal events reveals a significant temporal shift in the competition calendar. In 2020, events were concentrated in the latter part of the year, which marked a clear departure from the traditional midsummer peak. This shift toward late summer and autumn remains partially visible in 2021, though the schedule shows a gradual return to the pre-pandemic seasonal norm.

3.2. Median Analysis

The Mann-Whitney-U test confirms a significant decline in performance levels across all athletic categories during the 2020 season compared to the 2015–2019 baseline ($p < 0.05$). The overall median IAAF score decreased by 2.57%, falling from 914.0 to 890.5 points. The most substantial impact was observed in the Throw category, which saw a 3.99% reduction in median performance. Jumps and Sprints followed with declines of 2.58% and 1.35%, respectively. In contrast, the Run category demonstrated the highest resilience, showing the smallest significant decrease of 0.88%.

3.3. Density-Analysis

The bootstrap analysis of the Interquartile Range (IQR) reveals significant shifts in performance dispersion during the 2020 season compared to the 2015–2019 baseline. The most pronounced effect was observed in the Sprint category, where dispersion rose by 12.93% (95% CI: $[3.42\%, 20.00\%]$), followed by the Run category with a significant increase of 6.41% (95% CI: $[0.65\%, 13.82\%]$). In contrast, both the Throw category (-0.18%; 95% CI: $[-6.99\%, 9.42\%]$) and the Jump category (-6.79%; 95% CI: $[-11.85\%, 4.58\%]$) showed no statistically significant changes in dispersion.

Figure 1 shows the change in performance spread and typical performance development from 2021-2025 compared to 2001-2005 in many groups. The analysis of the performance data encompasses 106 groups, where X groups were excluded for median calculations (see Section 2). Overall, a decrease in performance density is evident. The best performers are improving, while the weaker performers are declining. While median performance trends were nearly balanced with 38 groups significantly improving and 37 de-

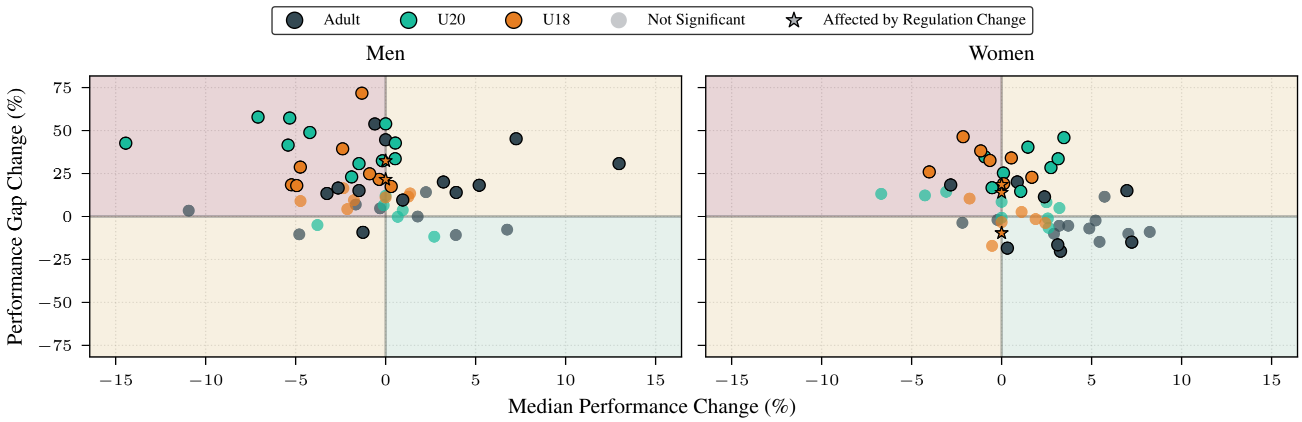

Figure 1. Relationship between median performance and change in performance gap (2001–2005 to 2021–2025). The scatter plot shows the percentage change in median performance (x-axis) and the change in performance gap (y-axis) for male (left) and female (right) groups, categorized by age group. The transparency of the markers indicates the statistical significance of the change in performance gap (via bootstrap method). Disciplines affected by regulatory changes are centered at 0 for the median, as only performance gap remains comparable across the time periods.

clining, the performance spread shows a strong asymmetry: A significant expansion was observed in 49.1% (52 groups), while a reduction was only seen in 4.7% (6 groups). This widening of the performance gap was most pronounced among men, especially in the U20 (70.6%) age group. In contrast, the women's groups showed significantly greater median progress: 63.3% improved compared to only 36.5% of the men's groups. Furthermore, discipline-specific trends emerged: The running and sprinting disciplines showed the most significant progress, with an improvement-to-decline ratio of 3:1 and 2:1 respectively. In contrast, the technical disciplines showed a significant decline: the declines outweighed the improvements by a ratio of 6:1 in throwing and 2.6:1 in jumping.

Figure 2 illustrates a linear polarization of performance. While the top-ranked company improved by an average of 1.61%, the 35th-rank declined by 1.51%, with the turning point towards negative growth being reached at the 22nd-rank. This polarization is not apparent only among adult women; all ranks from third place onward improved more than the top two ranks. Overall, women in all age groups perform better than their male counterparts. The most significant declines were seen in the male age groups under 18 and under 20. In the under-18 age group, the decline begins as early as rank 6, while in the under-20 age group it starts as early as rank 9.

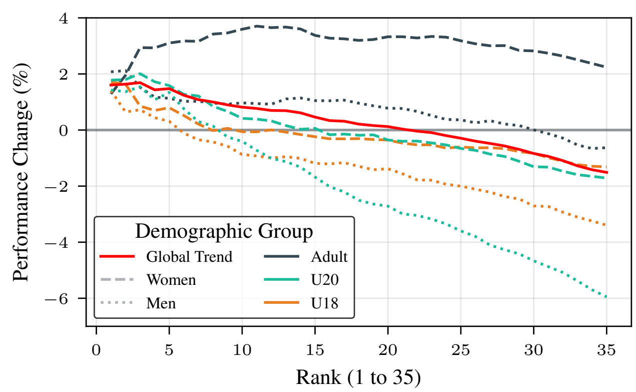

Figure 2. Performance development across ranks 1–35 (2001–05 to 2021–25). Shown is the percentage change in the average performance values of all groups overall, as well as broken down by age group and gender.

4. Discussion & Conclusion

Discussion on Performance Spread

Our results confirm a systematic increase in the performance gap, which aligns with the observed decline in performance at national championships. This reduction in performance density is particularly evident among men and young athletes, where performance differences have widened significantly. A major reason for this structural change is likely the 11.9% decrease in German Athletics Association (DLV) membership, exacerbated by a 26.6% drop in the 27- to 60-year-old age group (Lei). Since this age group contains a large number of coaches, its decline directly impacts technical disciplines that rely heavily on expert, individualized training. In contrast, women's athletics shows more improvements, which is probably due to a "catch-up effect" as a result of historically delayed professionalization.

Ultimately, the data reveals a polarization of performance: while professionalization protects the national elite, the second tier is shrinking. This confirms our hypothesis that the lack of internal pressure—true to the principle that "competition promotes performance"—has isolated the German elite and ill-prepared them for the intensity of international competition. The zero medal results of recent years are therefore not anomalies, but the predictable outcome of a shrinking talent pool.

Übergang zur Analyse von Luca

In running competitions, the overall improvement in average performance was significantly influenced by the introduction of carbon fiber shoes in 2016 (Willwacher et al., 2023). Furthermore, the COVID-19 pandemic led to uneven development. Endurance athletes may benefited from focused, uninterrupted training blocks (Venkatesan, 2022). In contrast, athletes in technical disciplines were hindered by the closure of training facilities.

Limitations

One limitation is the focus on the top 35. We cannot make any statements about the general development in all of amateur sports. The analysis only considers advanced competitive sports. Whether the observed trend continues linearly would need to be investigated with further analyses.

Conclusion

The increase in the performance spread indicates a structural change in organized athletics. A future focus should be placed on strengthening the grassroots level. Following the principle of "the grassroots supporting the elite," this will allow for more top performances and could bring German athletics closer to the world's elite.

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